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Insights into structure and properties of cellulose nanofibrils (CNFs) prepared by screw extrusion and deep eutectic solvent permeation

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ABSTRACT

To achieve the balance on economy and ecology, it is indispensable to explore the greener and more inexpensive method for the production of cellulose nanofibrils (CNFs). Herein, a deep eutectic solvent (DES) system based on choline chloride (ChCl) and ethylene glycol (EG) was employed as the swollen solvent, combining with screw extrusion and permeant, to fabricate unmodified CNFs with high yield and thermal stability. The proposed method in this work was simple, convenient, and industrially viable.

The hydrous DESs were applied in the process of CNFs preparation and dispersion to reduce the cost and viscosity of DES. To reveal the principle of CNFs preparation, the impact of sulfuric acid and water content of DES system on the chemical, physical, morphological, thermal, and dispersive properties of CNFs was systematically studied. Properties of the dispersed solvents were characterized by solvatochromic parameters and viscosity parameters to evaluate the potential influence on the preparation and dispersion of CNFs. In general, this work would play valuable guidance in realizing the preparation and dispersion of CNFs via a versatile DES solvent system, thus endowing cellulose materials high-value utilization.

1. Introduction

Cellulose, as an inexhaustible and widespread natural and green polymer (annual production capacity was around 75-100 billion tons worldwide), had attracted wide of increasing concern because of the global warming and depletion of petroleum sources [1,2]. Cellulose nanofibrils (CNFs), a kind of renewable and biodegradable cellulose nanomaterials [3,4], had received much attention in recent years because of the excellent features such as high strength and stiffness, high specific surface area, low density, biodegradability, and renewability. These advantages suggested that CNFs had great potential application prospect in composites to replace plastics [5,6], gel materials [7,8], catalysis [9,10], adsorption [11,12], electronics [13,14], and biomedical systems [15]. However, the structural complexity of components in natural cell walls limited the productivity of CNFs. Therefore, more efforts should be devoted to developing a highly efficient and eco-friendly method for CNFs preparation, which was conducive to reap the full potential value of cellulose nanomaterials in the future.

Cellulose fibrils contained crystalline and amorphous regions and it could be delaminated to CNFs via mechanical and mechanochemical methods. It was well known that the structure of cellulose fibrils was complex and tightly-packed formed by hydrogen bonds and van der Waals forces. Therefore, the mechanical treatment could defibrillate the fibers and transformed to CNFs as the mechanical energy exceeded the interfibrillar bond energy. Common methods for mechanical production of CNFs included high-pressure homogenization [16], ball milling [12], grinding [17] and so on.

However, mechanical disintegration for production of CNFs was of extensive energy consumption, low efficiency, and high cost because of the highly ordered hydrogen bond and dense network structure of

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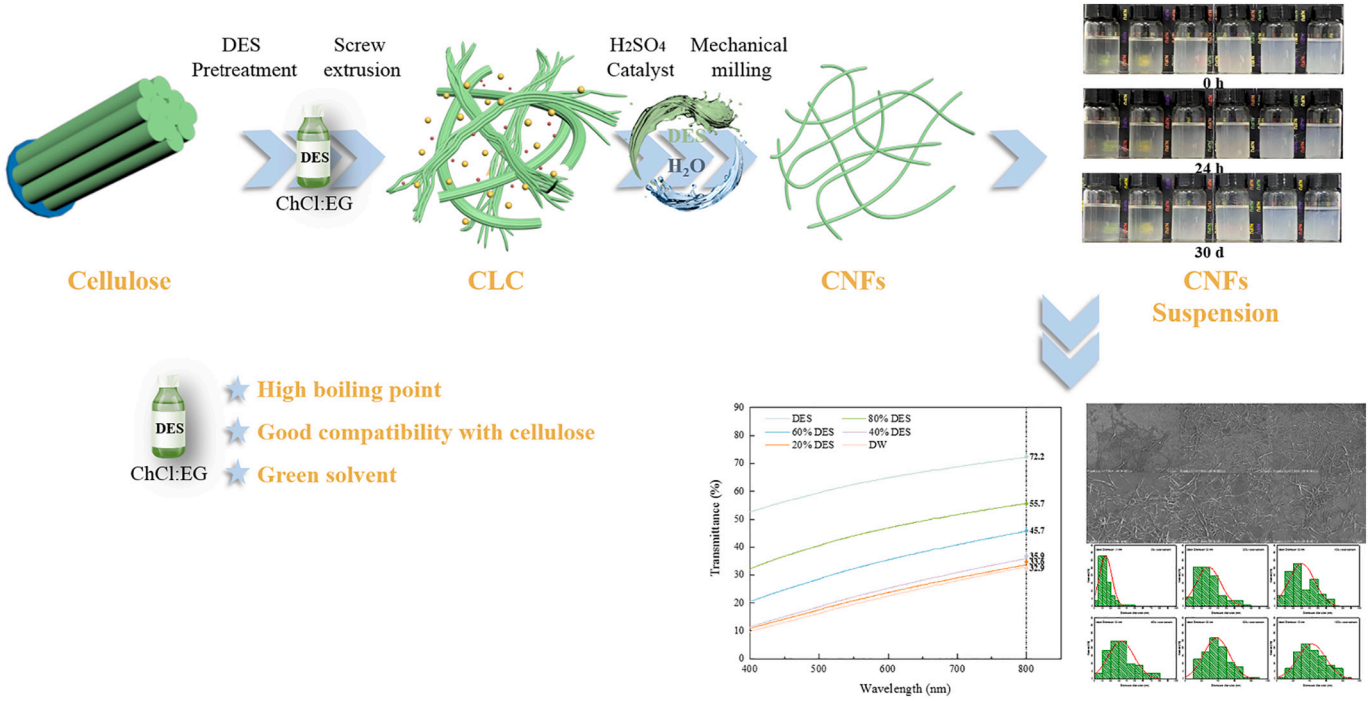


Fig. 1. A flowchart schematic of experiments for preparing CNFs.

cellulose, thus hindering the industrialized production of CNFs [18]. Scientists gradually recognized the problem and have found some new approaches to compensate for the shortcomings of single mechanical treatment for CNFs preparation. Chemical or biological pretreatment could swell cellulose fibers and partially disrupt the hydrogen bond network of cellulose fibrils, thus increasing the accessibility of cellulose fibrils [19]. That is, compared to single mechanical treatment, combining biochemical pretreatments with mechanical action could be conducive to reduce the energy consumption and improve the efficiency of fibrillation during the process of CNFs preparation. Thus far, various kinds of biochemical pretreatments have been used in CNFs preparation, such as acid hydrolysis, oxidation reaction, enzymatic hydrolysis, and so on. In the pretreatment of acid hydrolysis, the representative species mainly contained mineral acids (like sulfuric acid, hydrochloric acid, and phosphoric acid) and organic acids (like p-toluenesulfonic acid, oxalic acid, and maleic acid). It was well known that the acid hydrolysis could remove the amorphous of cellulose and partially disrupt the hydrogen bond network of cellulose fibrils, thus benefiting the liberation of CNFs via various mechanical treatments [20]. Oxidation reaction of cellulose mainly consisted of 2,2,6,6-tetramethylpiperidine-1-oxyl radical (TEMPO)-mediated oxidation and periodate–chlorite oxidation. Oxidation reaction could selectively oxidize the primary hydroxyl groups to weaken inter-fibrillar hydrogen bonds of cellulose, which would facilitate the extraction of CNFs [21]. However, the main obstacles during the process of CNFs preparation such as the harsh chemical consumption, corrosion of the equipment, and impacting environment still existed. The principle of enzymatic pretreatment on cellulose was similar to acid hydrolysis. However, enzymatic pretreatment was milder and more eco-friendly compared with acid hydrolysis or oxidation reaction [22]. Unfortunately, there were still some drawbacks in enzymatic pretreatment such as long reaction time, high operation cost, and so on. Therefore, to achieve the balance on economy and ecology, it is indispensable to explore a greener, simpler, and more inexpensive method for the pretreatment of cellulose.

Deep eutectic solvents (DESs), as new green solvents with biodegradability and eco-friendly nature, were simply synthesized by mixing hydrogen-bond donors (HBD, such as quaternary ammonium salts) with hydrogen-bond acceptors (HBA, such as amines, amides, carboxylic acids, polyol) [23]. This meant that the properties of DESs could be flexibly designable through adjusting the HBA and HBD types and molar ratios to meet practical application needs. Apart from the above, DESs possessed many distinct physicochemical properties similar to ionic liquids, such as low melting points, negligible vapor pressure, high thermal stability, and nonflammability [24]. It was known that ionic liquids were recognized as good solvents for treatment of cellulose [25]. Thus, more and more researches employed DESs as the swollen solvents to pretreat the biomass to produce CNFs. Liu et al. applied an acidic DES based on choline chloride and citric acid in the pretreatment of bleached softwood kraft pulp and combined with high-pressure homogenize to prepare CNFs [26]. Hong et al. selected the DES composed of choline chloride and oxalic acid to swell the luffa sponge and adopted a two-step ultrasonication treatment to prepare elongated CNFs [27]. Unfortunately, DESs employed as swollen solvents in the process of CNFs preparation were commonly acidic DESs in most researches, which would cause the side reactions like esterification between acidic DESs and CNFs. More than this, it was known that high viscosity of DESs was a key problem, which would cause mass transfer problems and low dissolution mass concentration [28]. High cost of DESs was another key concern hindering the widespread application of DESs. Adding water to the DES system was recognized as a simple, efficient and low-cost method to tailor the properties of DESs, which could reduce the viscosity and cost of DESs [29]. To our knowledge, there were still few studies on the hydrated natural DESs applied in the preparation and dispersion of unmodified CNFs remained among the gaps.

Herein, we aimed to use a natural DES as the swollen solvent followed by extrusion and permeation to prepare unmodified CNFs with high yield and thermal stability. The proposed method in this work was simple, convenient, and industrially viable. Moreover, to reduce the cost and viscosity of DES, the hydrous DESs were innovatively applied in the process of CNFs preparation and dispersion. Ethylene glycol (EG) was recognized as one of common biomass-derived polyols and showed a high boiling point and fascinating compatibility with biomass. In this work, EG was employed as HBD and choline chloride as HBA to synthesize the natural DES, which was used as a well-swollen solvent to pretreat cellulose under high temperature. Then, the swollen cellulose fibers were crushed and permeated using a screw extrusion device. After

a small quantity of sulfuric acid catalysis, the obtained samples in different dispersive solvents (DES, hydrous DESs, and aqueous phase) were further fibrillated by mechanical milling (Fig. 1). To reveal the principle of CNFs preparation and dispersion, the impact of sulfuric acid and water content on the chemical, physical, morphological, thermal, and dispersive properties of CNFs was systematically studied. Moreover, physicochemical properties (e.g., viscosity, acidity, basicity, and polarity) of DES were denoted using solvatochromic parameters including E_{NR}, Kamlet−Taft parameters (K-T parameters) (e.g., π*, α, and β) values [30,31], and viscosity parameters. Then, the potential relationships between these parameters of DES and the preparation of CNFs were evaluated. The results provided insights into chemical, physical, and structural changes of CNFs in response to varied quantities of sulfuric acid and water content of DES. This work would play valuable guidance in realizing the preparation and dispersion of CNFs using a DES solvent system.

2. Material and methods

2.1. Materials

The Needle Bleached Kraft Pulp (NBKP) was provided from Jiangsu Province of China. Sulfuric acid (H_2SO_4, 98%), Ethylene glycol ($C_2H_6O_2$, 98%), Choline chloride ($C_5H_{14}ClNO$, 98%) were purchased from Shanghai Aladdin Bio-Chem Technology Co. Ltd., China. Cuprietheylenediamine solution ($C_2H_{18}N_4O_2Cu$) was purchased from China National Pulp and Paper Research Institute. All chemicals were used as received without any purification.

2.2. Methods

The uniform and clear DES was synthesized by ChCl and EG at a molar of 1:1 under 100 °C for 1 h. Four hydrated DESs were obtained by mixing the DES with water at the mass ratio of 80:20, 60:40, 40:60, and 20:80, which were referred to as 80% DES, 60% DES, 40% DES, and 20% DES, respectively. The NBKP of oven-dry weight was torn into small pieces and pretreated by DES at a mass ratio of 1:8 (120 °C, 120 min) with mechanical stirring for sufficiently swelling. After this procedure, the crushed cellulose (CC) was obtained using a screw extrusion device (CZR-2014, Guangzhou Chunziran Co. Ltd., China) 3 cycles under 100 °C. The type of screw extrusion was a single screw with L/D ratio of 20 and rotate speed of 55 rpm. 0%, 0.2%, 0.4%, 0.6%, and 0.8% w/w H_2SO_4 (based on the CC dispersion system) as a catalyst was separately dripped into the CC dispersion system, which was diluted to 5% solid content. The CC dispersion system was fully swelled at 100 °C for 0.5 h to insure that H_2SO_4 was permeated into the CC dispersion system. The obtained samples with the optimized H_2SO_4 amount were directly diluted to 0.8% concentration using DES, 80% DES, 60% DES, 40% DES, 20% DES, and distilled water (DW), respectively. Then the diluted samples were refined at 6000 rpm for 40 min under ambient temperature using a colloid mill with the narrow gap between the rotor and the stator was 0.1–3.0 mm (JIM-150, Wenzhou Kangding Machinery co. Ltd., China). Finally, before freeze-drying for later evaluation, the obtained different CNFs samples were dialyzed in DW for 7 days to completely remove DES. The obtained different CNFs samples were named as DES-CNF, 20% DES-CNF, 40% DES-CNF, 60% DES-CNF, 80% DES-CNF, and DW-CNF, respectively.

2.3. Characterizations

The yield of CNFs was calculated from the equation as follow:

$$Y(\%) = \frac{w_1}{w_2} \times 100\% \quad (1)$$

where Y was defined as the yield of CNFs, W_1 was defined as the freeze-dried weight of.

CNFs (after dialysis), and W_2 was defined as the initial weight of raw material (oven dry).

A JEOL-JEM 7600F scanning electron microscope (SEM) was employed to observe the morphologies of CNFs samples. CNFs samples were prepared from CNFs suspensions with a concentration of 0.01 wt% and deposited on a silicon wafer and air-dried. The distribution of fibril diameter was measured using ImageJ software.

The degrees of polymerization (DP_v) of raw material and CNFs were measured using an Ubbelohde viscometer based on the Mark-Houwink-Sakurada equation [32].

$$DPv^{0.905} = 0.75[\eta] \quad (2)$$

The X-ray diffraction (XRD) of CNFs samples was measured by an X-ray refractometer (Ultima IV, Japan) with CuKa radiation (40 kV and 30 mA) in the range from 5° to 50° at a scan rate of 2°/min. The crystallinity degree (C*r*) of CNFs samples was calculated based on the Segal method [33]:

$$Cr = \frac{I_{002} - I_{am}}{I_{002}} \times 100\% \quad (3)$$

The Fourier transform infrared spectroscopy (FTIR) spectrogram of CNFs samples was made using a Fourier-transform infrared spectrometer (Vertex 80 V Bruker Germany) in a spectral range of 4000–450 cm^{-1} with a resolution of 2 cm^{-1}.

The thermogravimetric analyses (TGA) of NBKP and CNFs samples were conducted on a thermogravimetric analyzer (NETZSCH, TGA 209F1, Germany). CNFs Samples of 6-10 mg were evaluated from 30 °C to 600 °C at a heating rate of 10 °C/min under a 20 mL/min nitrogen stream.

DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF were dispersed in DES, 80% DES, 60% DES, 40% DES, 20% DES, and DW at a concentration of 0.1 wt% to observe the dispersion of CNFs samples, respectively. The obtained CNFs dispersions were sonicated in a bath sonicator (Avantech AC220, 600 W, 40 kHz) for 20 min and then evaluated their transmittance using a TU-1900 ultraviolet-visible spectrophotometer (UV–vis) in the range of the wavelength of visible light.

The solvatochromic properties [31] of DES and hydrated DESs were characterized by solvatochromic parameters including E_{NR} and *K-T* parameters (π*, α, and β) values according to the method reported in the literature. E_{NR} was quantified the polarity of solvents and Kamlet–Taft parameters (K-T parameters) including α, β, and π* were quantified hydrogen-bond donating ability (acidity), hydrogen-bond accepting ability (basicity), and polarity/polarizability, respectively. The viscosity of DES and hydrated DESs were tested using a digital viscosimeter (FANGRUI, NDJ-001, China). All samples were measured by three replicates and the averages were reported.

3. Results and discussion

3.1. CNFs preparation

The flowchart of CNFs preparation was schematically shown in Fig. 1. DESs, as emerging kinds of solvents synthesized by HBA and HBD, were much significant for the swelling of cellulose [34]. It was found that EG exhibited good compatibility with cellulose according to the previous literature [35].

Moreover, EG was one of the biomass-derived polyols and it possessed a high boiling point. Therefore, DES consisted of ChCl and EG could promote cellulose fibrillation under high temperature, due to the strong interaction with cellulose chains. Based on these findings and previous studies by our group [36,37], DES consisted of ChCl and EG was chosen as a swollen solvent to pretreat cellulose and yielded CNFs in this work. First, DES was employed as a reaction medium to soften, permeate, and swell the NBKP pieces at 120 °C for 2 h. The pretreatment

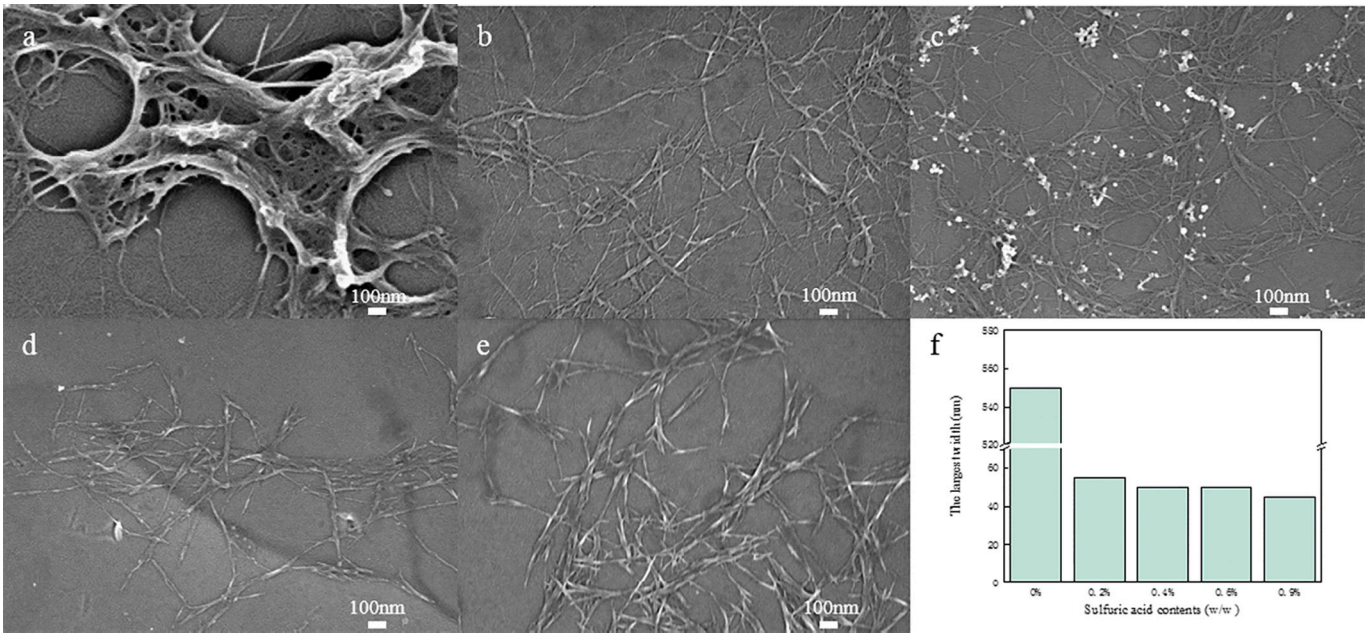


Fig. 2. The morphological images of the prepared CNFs with different H_2SO_4 amounts (a, 0%; b, 0.2%; c, 0.4%; d, 0.6%; e, 0.8% w/w) and (f) the largest width of the prepared CNFs.

temperature of 120 °C was a suitable temperature to swell NBKP pieces for promoting cellulose fibrillation [36,38]. Completely DES permeation and swelling of cellulose fibers was not only in favor of increasing the reactive activity but also conducive to the subsequent extrusion process. Second, the fully softened, permeated, and swollen cellulose was crushed using the screw extruder, which combined kneading and feeding screws. Due to the severe mechanical shear force and interaction friction, the fully swollen cellulose fibers were disrupted, torn, and then cut off to finer fibers. It was proved that the cell wall of cellulose fibers could be disintegrated into elementary fibrils during the screw extrusion [39]. Namely, more DES could be absorbed by crushed cellulose fibers under the action of mechanical extrusion. Trigui et al. reported that the swollen cellulose fibers could be successfully disintegrated into CNFs by twin-screw extrusion [40]. Then, the CC samples were diluted to 5% solid content by DES and further.

hydrolyzed under 100 °C for 0.5 h using 0%, 0.2%, 0.4%, 0.6%, and 0.8% w/w H_2SO_4 (based on the CC dispersion system) as a catalyst, respectively. Due to the catalysis of H_2SO_4, cellulose fibrils were more accessible to hydrolyze and the cleavage of linkages among cellulose fibrils was partly broken. The obtained CC dispersion systems with the optimized H_2SO_4 amount were diluted to 0.8% concentration using DES, 80% DES, 60% DES, 40% DES, 20% DES, and distilled water (DW), respectively, for evaluation of the water content of DES system affecting the preparation and dispersion of CNFs. Ultimately, all samples were delivered into a colloid mill for further refining to prepare CNFs suspensions and the diameters of obtained CNFs were all less than 100 nm (Fig. 1). This phenomenon suggested that CNFs were successfully prepared under the synergism of DES swelling, screw extrusion, a small amount of sulfuric acid catalysis, and mechanical milling.

3.2. The impact of dosages of sulfuric acid on the preparation of CNFs

Fig. 2 presented the morphological images of as prepared CNFs using different quantities of H_2SO_4. It could be observed that nano-sized cellulose fibrils were not completely obtained without H_2SO_4 catalysis, although the cellulose fibrils were torn, destroyed and delaminated by DES swelling and mechanical action (Fig. 2a). In contrast, the cellulose fibrils could be successfully disintegrated into CNFs with H_2SO_4 catalysis (Fig. 2b–c). Moreover, it could be drawn that the diameters of cellulose

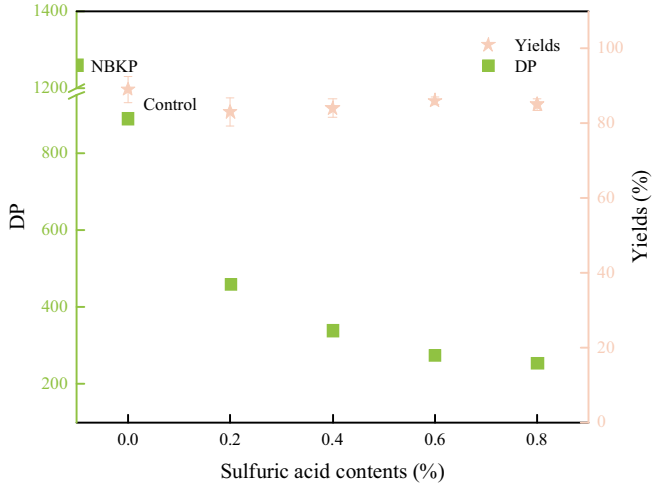


Fig. 3. The DP_v and yields of CNFs with different H_2SO_4 amounts and NBKP.

fibrils were finer with the increase of H_2SO_4 amounts (Fig. 2f). As was well known that, H_2SO_4 as a catalyst could hydrolyze and activate cellulose to facilitate the subsequent mechanical action to the liberation of nano-sized cellulose fibrils.

To further explore the role of small quantities of H_2SO_4 in the process of CNFs preparation, the yields and DP_v of CNFs with different quantities of H_2SO_4 were evaluated (Fig. 3). It visibly revealed that the yields of CNFs using different quantities of H_2SO_4 catalysis ranged from 83% to 86%, while the control sample (without H_2SO_4 catalyst) was 89%. Moreover, no notable changes in CNFs yields were observed as the increase of H_2SO_4 amount. This phenomenon proved that small quantities of H_2SO_4 could slight hydrolysis and swelling cellulose fibrils and benefit for the preparation of CNFs, while did not cause intensive hydrolysis. The decrease of the CNFs yields might be caused by two following reasons: (1) some samples remain in the screw extruder or colloid mill due to the experimentally operating loss; (2) the hydrolysis of the amorphous section of cellulose fibrils. The DP_v of samples were decreased with the increase of H_2SO_4 amount (Fig. 3), which was

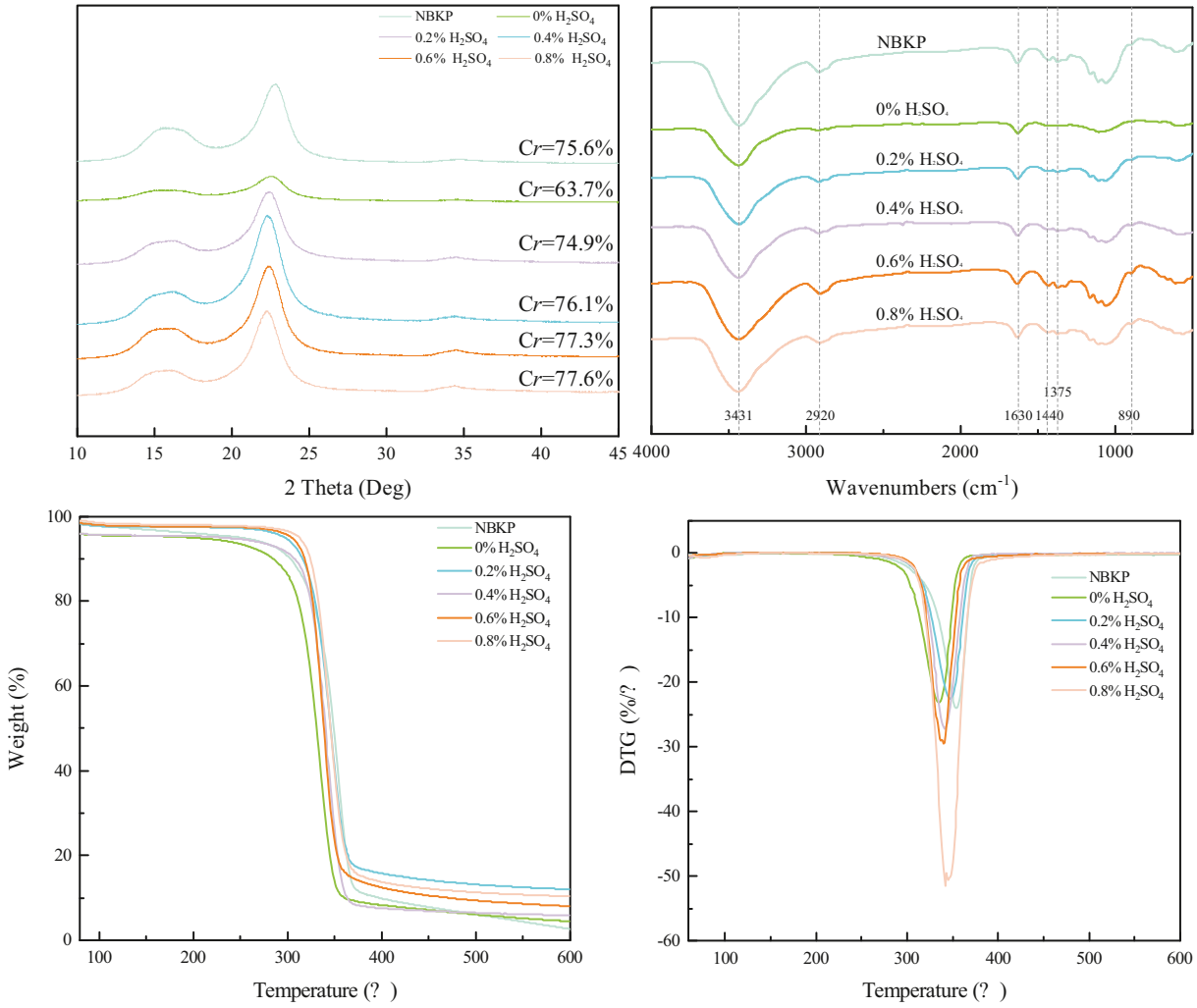


Fig. 4. The XRD, FTIR, TGA, and DTG of CNFs with different quantities of H_2SO_4 and NBKP.

different from the variation of CNFs yields. The DP_v of NBKP was 1260 and the control sample (without the addition of H_2SO_4) was 890, which indicated that DES swelling and mechanical action could effectively reduce the DP_v of cellulose [41,42]. When the H_2SO_4 amount increased from 0.2% to 0.8%, the DP_v of samples was decreased from 460 to 255. These results indicated that a small amount of H_2SO_4 could slightly hydrolyze cellulose fibrils, which played a significant role in the preparation of CNFs.

XRD diffraction patterns of obtained CNFs and NBKP were shown in Fig. 4. It was obviously indicated that the crystalline structure of NBKP and all CNFs samples were the cellulose I structure, as the XRD curves of obtained CNFs and NBKP were similar to previous literatures [43]. Thus, we could conclude that the crystal structure of obtained CNFs was hardly changed by DES swelling, mechanical treatment, and H_2SO_4 catalysis. The degrees of crystallinity computed from the 2θ peaks for the raw material and CNFs using different amounts of H_2SO_4 catalysis were 75.6%, 63.7%, 74.9%, 76.1%, 77.3%, and 77.6%, respectively. The crystalline index of CNFs without H_2SO_4 catalysis was the lowest among all samples. It was probably due to the mechanical action, which could destruct the crystalline structure of samples, thus reducing the crystalline index [17]. When H_2SO_4 catalysis was employed, the crystalline index of CNFs increased compared with the raw material. Moreover, it could be found that the crystalline index of CNFs became larger as the usage of H_2SO_4 increased. Combining with the results in our previous studies [36,37], the phenomenon were likely caused by more dissolution of the amorphous region under the action of acid hydrolysis.

Table 1
Thermal degradation behavior of CNFs using different quantities of H_2SO_4 and NBKP.

Samples	T_{onset} (°C)	T_{max} (°C)	Residue (%)
NBKP	330	354	2.68
DES-CNF (0% H_2SO_4)	311	334	4.49
DES-CNF (0.2% H_2SO_4)	324	344	9.25
DES-CNF (0.4% H_2SO_4)	324	341	12.02
DES-CNF (0.6% H_2SO_4)	323	340	5.89
DES-CNF (0.8% H_2SO_4)	335	342	8.04

To study the changes of the chemical structure during the process of CNFs, the FTIR of NBKP and all CNFs were investigated. As shown in Fig. 4, the dominant peak at 3431 cm^{-1} and 1630 $cm^{-1}$was due to -OH stretching vibration and O—H bending vibration, respectively [44,45]. The 2920 cm^{-1} band was referred to C—H stretching vibration and the 1440 cm^{-1} peak was attributed to intermolecular hydrogen attraction at C6 group [46]. The bands at 1375 cm^{-1} belonged to C—H and C—O bonds in the polysaccharide aromatic ring. The 890 cm^{-1} peak was referred to β-glycosidic linkages between glucose units in cellulose [47]. Overall, the distinct peaks in the FTIR spectra of CNFs were parallel to the natural NBKP, indicating that the preparation of CNFs did not change the main chemical structure [34]. Thus, it could be confirmed that the few amount of H_2SO_4 was employed as a catalyst rather than a modifier in the preparation of CNFs.

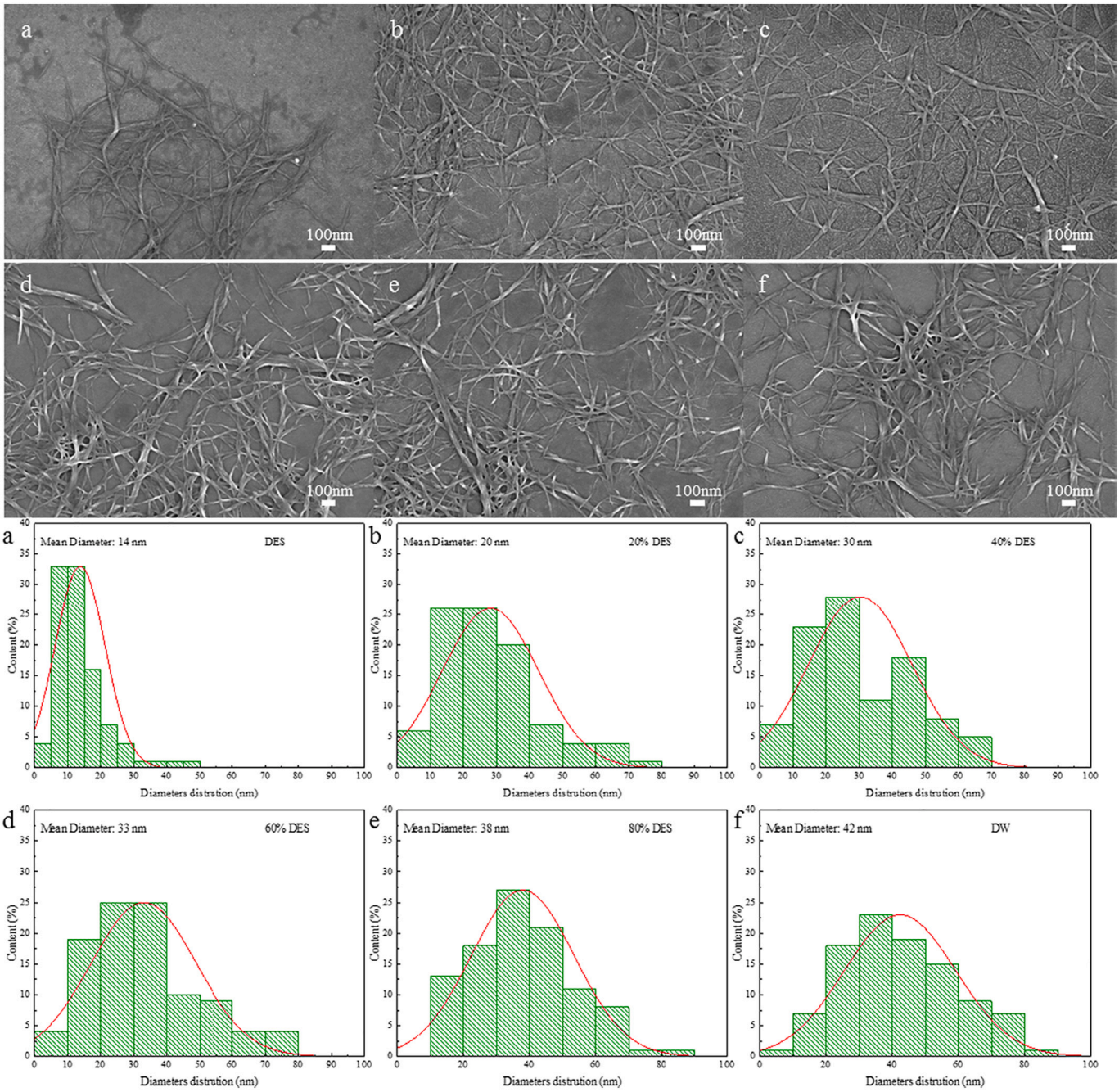


Fig. 5. The morphological images and diameters distribution of corresponding CNFs (a, DES-CNF; b, 80% DES-CNF; c, 60% DES-CNF; d, 40% DES-CNF; e, 20% DES-CNF; f, DW-CNF).

Fig. 4 showed the thermal stability and degradation characteristics of NBKP and CNFs using different quantities of H_2SO_4, which were investigated by TG and DTG. The initial temperature (T_{onset}), the maximum temperature of decomposition (T_{max}), and the residue mass were summarized in Table 1. The first degradation region ranging from 30 to 120 °C was associated with the dehydration of CNFs or NBKP. As shown in Fig. 4 and Table 1, the T_{onset} of samples was 330 °C, 311 °C, 324 °C, 324 °C, 323 °C, and 335 °C, respectively for NBKP, DES-CNF (0% H_2SO_4), DES-CNF (0.2% H_2SO_4), DES-CNF (0.4% H_2SO_4), DES-CNF (0.6% H_2SO_4), and CNFs (0.8% H_2SO_4). It could be found that the T_{onset} of all DES-CNF was lower than the raw material in addition to DES-CNF (0.8% H_2SO_4). The reasons for the high T_{onset} could be speculated as follows [34,48]: (1) the higher crystalline index and (2) the removal of more amorphous regions. However, it was worth noting that the thermal stability of obtained DES-CNF was higher than the traditional acid hydrolyzed CNFs and acid-DES prepared CNFs [20].

3.3. The impact of hydrated DESs on the preparation of CNFs

After a series of experiments and analyses on different dosages of H_2SO_4 affecting on the preparation of CNFs, the optimal dosage of H_2SO_4 was 0.4% w/w (based on the CC dispersion system). The main reason was that when the dosage of H_2SO_4 reached up to 0.4% w/w, the diameter of CNFs was fine and uniform. In the meantime, the yield and DP_v of DES-CNF (0.4% H_2SO_4) were exhibited well. Therefore, 0.4% w/w H_2SO_4 was chosen as a catalyst, combining with DES swelling and screw extrusion to pretreat cellulose fibrils. Then, the obtained samples were diluted to 0.8% w/w concentration using DES, 80% DES, 60% DES,

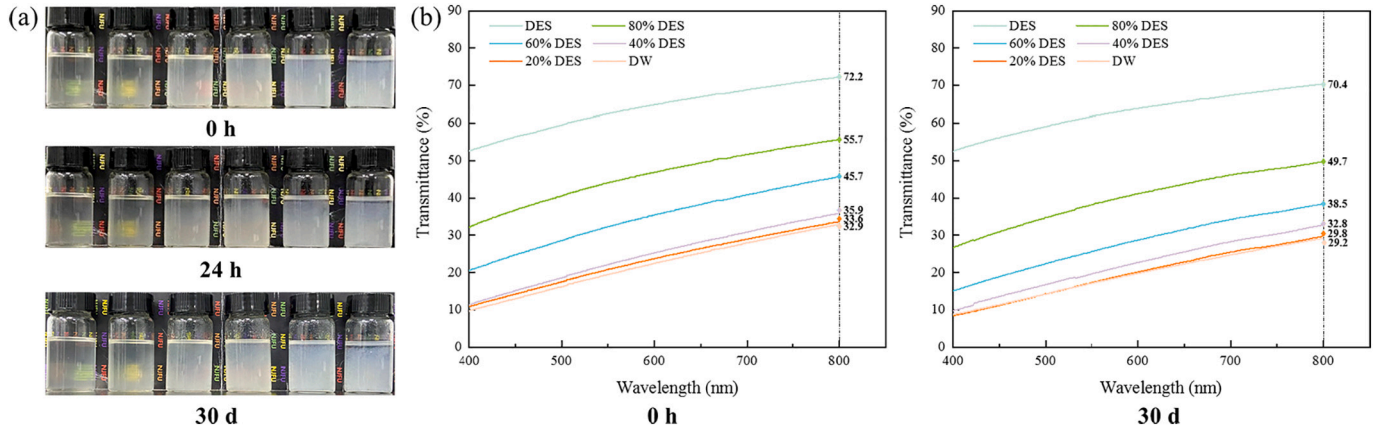


Fig. 6. (a) The photographs of the dispersion of 0.1% CNFs suspensions in DES, 80% DES, 60% DES, 40% DES, 20% DES, and DW after 0 h (hours), 24 h, and 30 d (days): a, DES-CNF; b, 80% DES-CNF; c, 60% DES-CNF; d 40% DES-CNF; e, 20% DES-CNF; f, DW-CNF. (b) The UV–vis spectra of CNFs in DES, 80% DES, 60% DES, 40% DES, 20% DES, and DW.

Table 2
Solvatochromic parameters and viscosity of DES, 80% DES, 60% DES, 40% DES, 20% DES, and DW.

Solvent	E_{NR}	π*	α	β	Viscosity (mPa s)
DES	50.51	0.98	0.80	0.71	42.1
80% DES	49.29	1.17	0.95	0.49	14.2
60% DES	48.54	1.24	1.07	0.41	6.8
40% DES	48.21	1.34	1.08	0.24	4.1
20% DES	47.57	1.34	1.22	0.19	2.8
DW	/	1.34	1.23	0.14	/

40% DES, 20% DES, and DW, respectively. In the end, the samples were separated to nano-scaled fibers by a colloid mill. In this procedure, the morphologies and diameter distributions of obtained CNFs using different dispersive solvents would be evaluated by SEM. As shown in Fig. 5, CNFs were successfully prepared using DES, hydrated DESs, and DW as dispersive solvents. However, the microtopographies of obtained CNFs using different dispersive solvents were different. It could be found that the entangled state of CNFs was more distinct as the increase of water content of dispersive solvents. Correspondingly, the mean diameter of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF was 14, 20, 30, 33, 38, and 42 nm, respectively. These results made us conclude that the diameter distribution of CNFs was narrower with the decrease of water content of dispersive solvents. It might be connected to the properties of different dispersive solutions, thus hindering the cellulose fibrils liberated to nanosized fibrils. Further discussion about solvation properties of dispersive solvents would be made below.

The photographs and light transmittances of CNFs suspensions were displayed in Fig. 6, reflecting the aggregation and dispersion of CNFs. It could be found that there was no apparent aggregation in DES-CNF, 80% DES-CNF, and 60% DES-CNF suspensions. While 40% DES-CNF, 20% DES-CNF, and DW-CNF suspensions showed more turbid compared with others, which might be resulted from the aggregation and large diameter of CNFs. Storing for long periods (30 days), no obvious sedimentation in DES-CNF and 80% DES-CNF suspensions could be observed, which were uniform and stable. Instead, the aggregation of 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF suspensions was clearly observed. The cause of this phenomenon could be explained by the compatibility principle, steric hindrance, and gravity in good balance [49]. From Table 2, it could be found that DES and 80% DES possessed high viscosity, indicating that they had well steric hindrance. Correspondingly, the dispersion of DES-CNF and 80% DES-CNF suspensions were more stable and uniform than others within 24 h. After storing for 30 days under the ambient temperature, the impact of steric hindrance on the CNFs suspensions would diminish. However, it was observed that DES-CNF and 80% DES-CNF suspensions remained stable and uniform after 30 days. It could be speculated that the steric hindrance had not considerable impact on CNFs suspensions after 30 days in this work. In this stage, the well compatibility between DES and CNFs hindering the reaggregation of CNFs in the dispersive solvent was plausibly the key factor. According to the previous literature [50], it was reported that the dispersion of CNFs suspensions could be significantly improved as the decrease of fibers' diameter. Combined with the SEM analysis (Fig. 5), we considered that the good compatibility between DES and cellulose was contributed to improving the swelling and accessibility of cellulose fibrils and facilitating mechanical action to disintegrate cellulose fibrils, thus making the CNFs diameter finer [51,52].

It was well known that the light transmittance was generally used as an indicator to evaluate the dispersion of CNFs suspensions as it related to the width and aggregation of CNFs [53]. The transmittance of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF suspensions was 72.2, 55.7, 45.7, 33.5, 30.8, and 29.2%, respectively (Fig. 6b). It suggested that the aggregation of CNFs would be more serious with the addition of the water content of dispersive solvents, which was corresponding to the result of photographs of CNFs suspensions. After 30 days, the light transmittance of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF suspension was 70.4, 49.7, 38.5, 32.8, 29.8, and 29.2%, respectively. Compared to the initial light transmittance, the light transmittance of DES-CNF and 80% DES-CNF suspensions only decreased 1.8% and 6%, respectively. It could support that the dispersion of DES-CNF and 80% DES-CNF were stable.

To further study the cellulose-solvent interactions, E_{NR} and K-T parameters (π*, α, and β) were used as tools to describe the properties of DES and hydrated DESs, which represented the polarity, dipolarity/polarizability, hydrogen-bond donating ability (acidity), hydrogen-bond accepting ability (basicity) of the solvents, respectively [30,31]. As well known, the polarity of solvents would decrease as the E_{NR} values raised. The solvation performance of K-T parameters (π*, α, and β) exhibited a direct proportion to the values. As listed in Table 1, the E_{NR}, π*, α, and β values of DES were 50.51, 0.98, 0.80, and 0.71, respectively. It was apparently found that the π* and α values were increased with the rise of the water content of dispersive solvents, while the E_{NR} and β values were decreased. Overall, it could be known that the basicity of DES was the highest, while the polarity and acidity was the worst among all dispersive solvents. The K-T parameters (π*, α, and β) of NMMO·H_2O, as a good solution for cellulose dissolution, was 1.067, 0.205, and 0.969, respectively [54]. Moreover, the literatures revealed that the polarity of EG was close to cellulose and the π* value was 1.04 [37,55]. Based on these

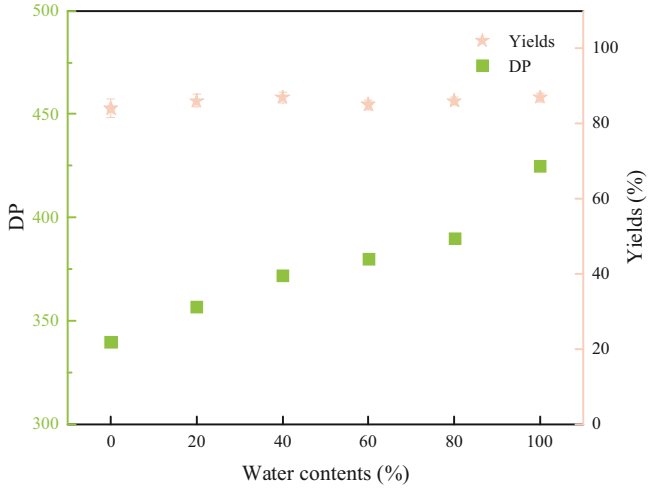


Fig. 7. The DP_v and yields of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF.

findings and the results in this research, it could be inferred that the π* value was a key parameter for cellulose-solvent interactions, determining the compatibility of solvent with cellulose. According to the results above, it could be concluded that well compatibility of solvents with cellulose would be conducive to the preparation and dispersion of CNFs. The viscosity of dispersive solvents was shown in Table 2. The viscosity of dispersive solvents decreased significantly with more addition of water content of dispersive solvents. In general, high viscosity could be considered as an adverse factor in the preparation of CNFs, which would cause mass transfer difficulty and hinder the liberation of nano-sized fibrils. However, the low viscosity of dispersive solvents did not promote the preparation of CNFs in this work. It could be speculated that the polarity of DW was too high to make the poor compatibility between DW and the cellulose, thus hindering the preparation and dispersion of CNFs.

As reflected in Fig. 7, the yields of CNFs prepared by 0.4% w/w H_2SO_4 and different dispersive solvents were ranged from 84–87%. Considering that the yield of micro-nano sized cellulose fibrils (89%) from raw material NBKP without acid catalyst, the yield of CNFs with different dispersive solvents was satisfactory. The DP_v of CNFs prepared by different dispersive solvents did not show a significantly decreased trend, ranging from 340 to 425. Megashah et al. proposed that shorter fibers might result in lower DP_v [56]. Moreover, the DP_v of CNFs would be decreased with the shorter length [42]. Therefore, it could be drawn that the DP_v of CNFs would increase as the water content of dispersive solvents raised, which was corresponding to the SEM and light transmittance analyses.

The chemical structure and thermostability of CNFs prepared by

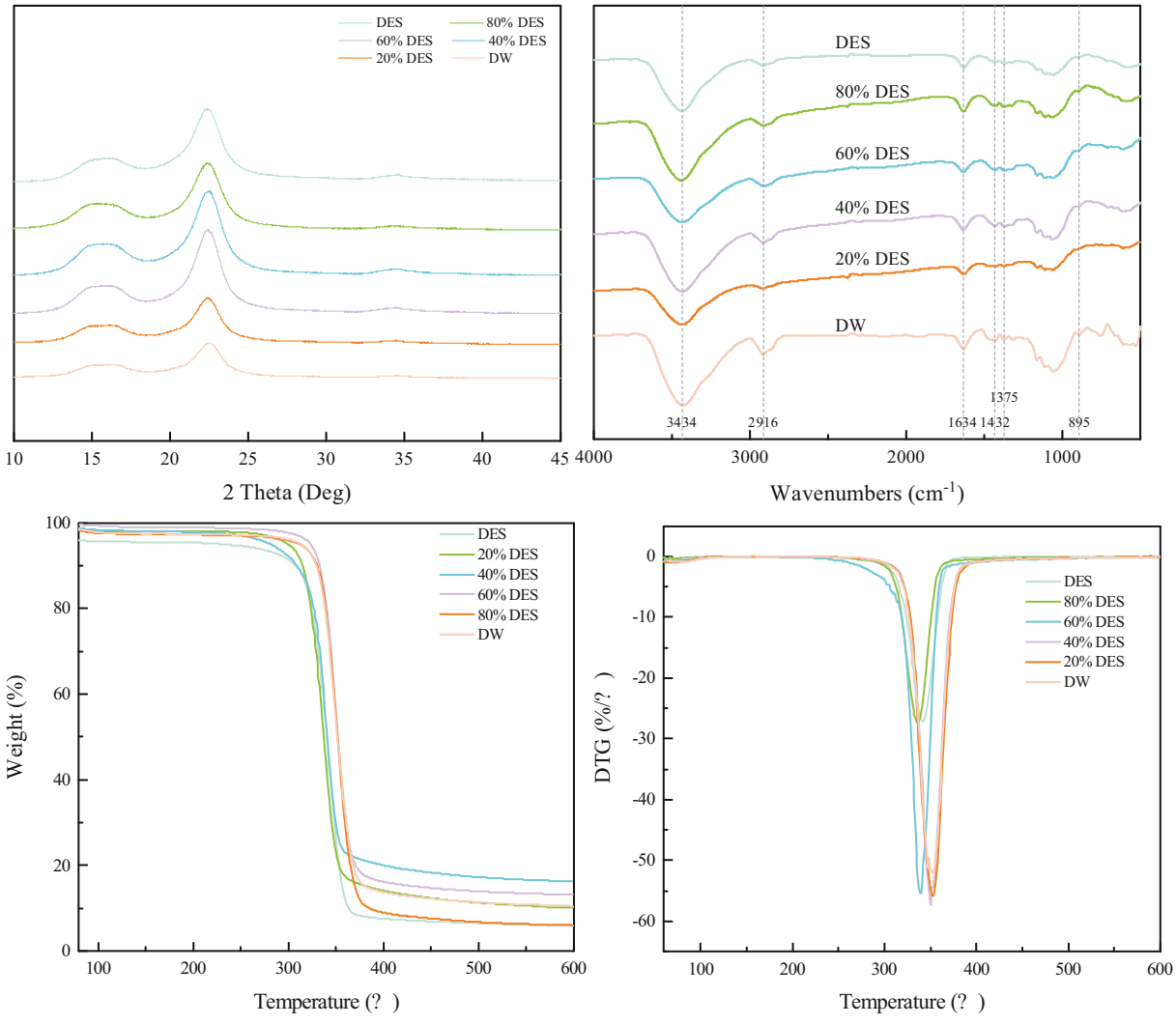


Fig. 8. The XRD, FTIR, TG, and DTG of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-CNF.

0.4% w/w H_2SO_4 and different dispersive solvents were characterized by XRD, FTIR, TG, and DTG (Fig. 8). The crystalline structure of all CNFs samples were shown in Fig. 8. Paralleling with the XRD patterns of CNFs prepared using diverse amounts of H_2SO_4 as a catalyst, the XRD diffractograms of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-DES showed characteristic peaks of cellulose I structure. The chemical structure of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-DES was investigated through the FTIR analysis. As seen from Fig. 8, the labeled peaks (3434, 2916, 1634, 1432, 1375, and 895 cm^{-1}) of all CNFs samples were similar to the raw material NBKP (Fig. 4), demonstrating that the surface chemical structure of CNFs remained unchanged. The TG and DTG curves of DES-CNF, 80% DES-CNF, 60% DES-CNF, 40% DES-CNF, 20% DES-CNF, and DW-DES were shown in Fig. 8. It was observed that all CNFs samples were exhibited good thermal stability, which was better than the CNFs obtained by acidic DESs [57,58].

4. Conclusions

High-yield and thermostable CNFs were successfully prepared via DES based on ChCl: EG swelling, screw extrusion and permeant, a minimum supply of sulfuric acid catalysis and colloidal milling, while preserving the natural chemical structure of CNFs. The hydrous DESs were innovatively applied in the process of CNFs preparation and dispersion to reduce the cost and viscosity of DES. To reveal the principle of CNFs preparation and dispersion, the impact of sulfuric acid and water content on the chemical, physical, morphological, thermal, and dispersive properties of CNFs was systematically studied. It was found that the properties of solvents (such as viscosity, solvatochromic properties) had a relationship with the preparation and dispersion of CNFs. More strikingly, the light transmittance of DES-CNF suspension was up to 72.2% and it could remain stable standing for 30 days, showing favorable dispersion and stability. In conclusion, this work, which was simple and industrially viable, would play valuable guidance on realizing the preparation and dispersion of CNFs via a versatile DES solvent system, thus making cellulose materials high-value utilization.

CRediT authorship contribution statement

Ming Yan: Conceptualization, Methodology, Writing-original draft preparation, Investigation, Writing-Reviewing and editing. **Chaochao Tian, Ting Wu, Xingyu Huang, Yidan Zhong, Pei Yang:** Experiment & analysis. **Lili Zhang:** Writing-original draft preparation, Writing-Reviewing and editing, Financial support. **JinXia Ma:** Methodology, Writing-Reviewing and editing. **Hailong Lu:** Methodology, Writing-Reviewing and editing, Supervision, Validation. **Xiaofan Zhou:** Methodology, Supervision.

Declaration of competing interest

The authors declare no conflict of interest.

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